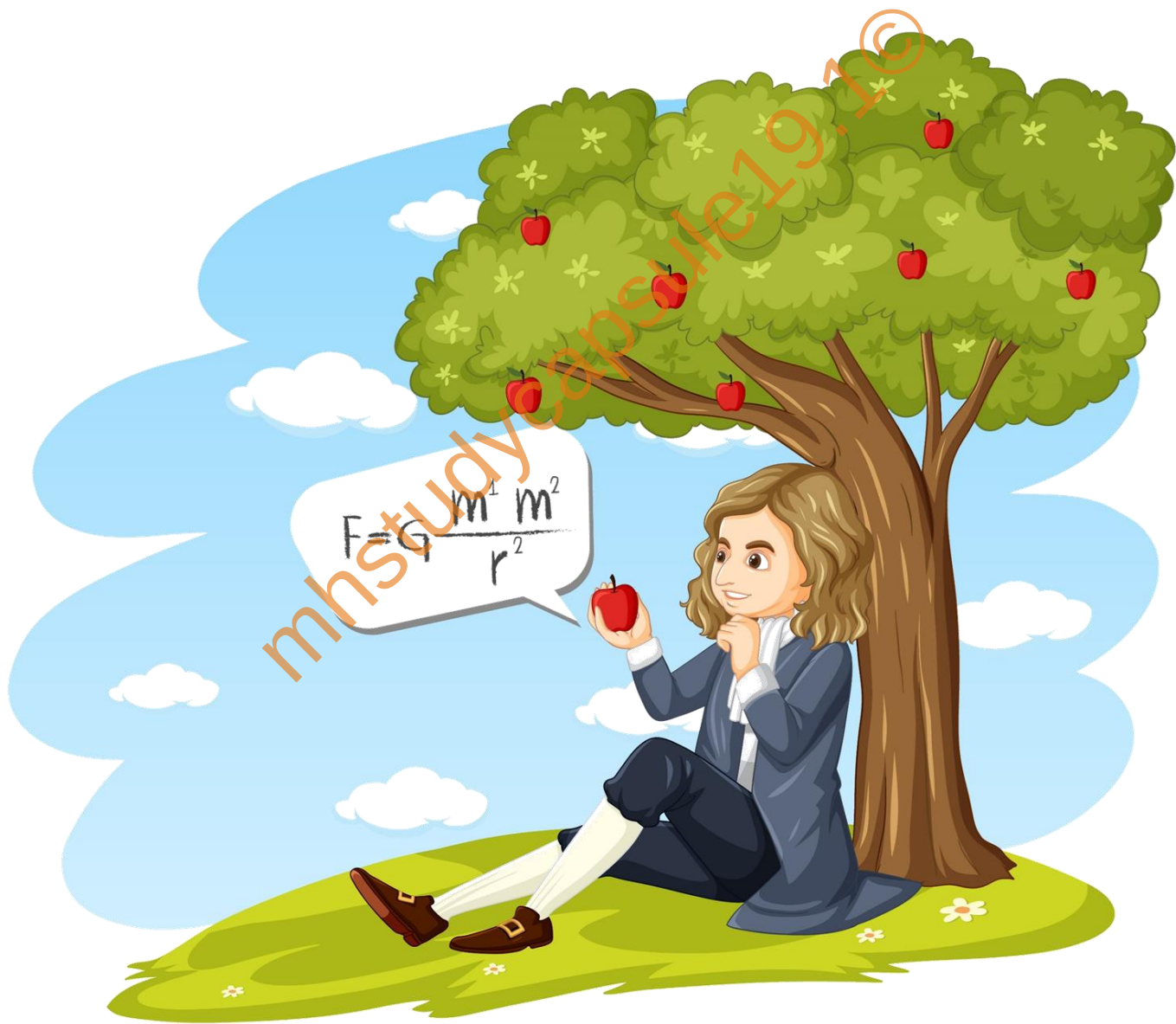


Chapter 1

Gravitation



Question And Answers

1. Study the entries in the following table and rewrite them putting the connected items in a single row.

I	II	III
Mass	kg	Measure of inertia
Weight	N	Depends on height
Acceleration due to gravity	m/s^2	Zero at the centre
Gravitational constant	Nm^2/kg^2	Same in the entire universe

2. Answer the following questions.

- a. What is the difference between mass and weight of an object.

MASS (m)	WEIGHT (W)
Mass is the amount of matter present in the object.	Weight is the force with which the Earth attracts the object.
Mass does not change with place.	Weight(W) = mg. As g changes, W also changes.
SI unit is kg. CGS unit is gram (g).	SI unit is newton. CGS unit is dyne.
Mass is a scalar quantity.	Weight is a vector quantity.

Will the mass and weight of an object on the earth be same as their values on Mars? Why?

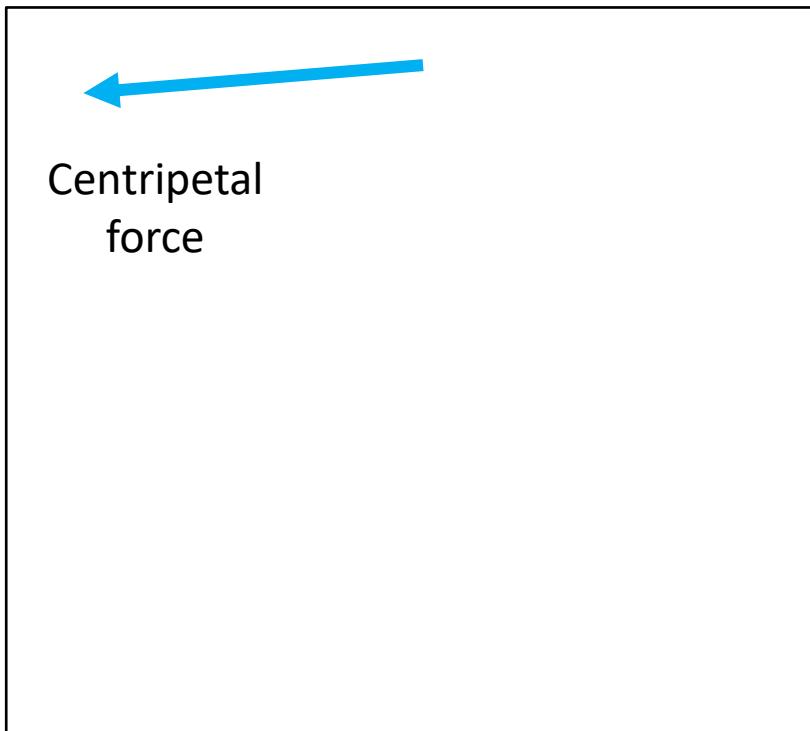
1. The mass of the object will remain same, but its weight will differ on Mars.
2. Since mass of an object is the amount of matter present in the body, its value remains the same on Earth and Mars.
3. Weight is the force with which the Earth attracts the object. It is given as $W = mg$. As g changes, weight (W) also changes.

b. What are (i) free fall, (ii) acceleration due to gravity (iii) escape velocity (iv) centripetal force ?

- Free fall:** When a body falls towards the Earth under the influence of the Earth's gravity alone, its motion is called a 'free fall'.
- Acceleration due to gravity:** The gravitational force due to earth on a body results in its acceleration. This is called acceleration due to gravity.
- Escape velocity:** The minimum initial velocity needed by an object projected upwards to overcome Earth's gravitational force and not fall back on Earth is called escape velocity.
- Centripetal force:** The force that acts on any object moving along a circle, directed towards the centre of the circle is called centripetal force.

SOMETHING TO REMEMBER

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$$F = \frac{mV^2}{R}$$

m = mass of object

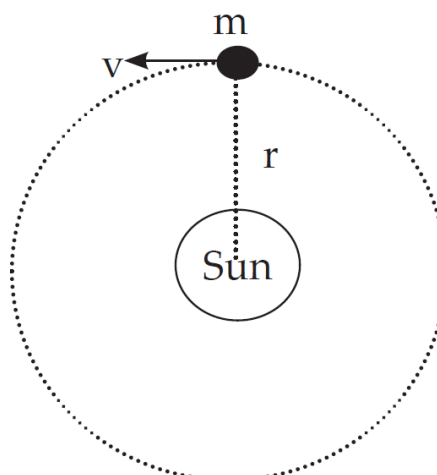
V = velocity of object

R = Radius of Circle

- c. Write the three laws given by Kepler. How did they help Newton to arrive at the inverse square law of gravity?

The three laws given by Kepler are as follows:

1. Kepler's first law: The orbit of a planet is an ellipse with the Sun at one of the foci.
2. Kepler's second law: The line joining the planet and the Sun sweeps equal areas in equal intervals of time.
3. Kepler's third law: The square of its period of revolution around the Sun is directly proportional to the cube of the mean distance of a planet from the Sun.



$$\text{Centripetal force } F = \frac{mv^2}{r} \quad \dots(i)$$

$$\begin{aligned}\text{Distance travelled by planet in one revolution} &= \text{Circumference of orbit} \\ &= 2\pi r\end{aligned}$$

$$\text{Time period of revolution} = (T)$$

$$\begin{aligned}\therefore \text{Speed of planet } V &= \frac{\text{distance}}{\text{time}} \\ &= \frac{2\pi r}{T} \quad \dots(ii)\end{aligned}$$

Substituting eq. (ii) in (i)

$$\therefore F = \frac{m\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{m4\pi^2 r^2}{rT^2}$$

$$\therefore F = \frac{4m\pi^2 r}{T^2}$$

Multiplying numerator and denominator with 'r²'.

$$\therefore F = \frac{4m\pi^2 r}{T^2} \times \frac{r^2}{r^2}$$

$$\therefore F = \frac{4m\pi^2 r^3}{r^2 T^2}$$

$$F = \frac{4 m \pi^2}{r^2} \times \left(\frac{r^3}{T^2} \right)$$

According to Kepler's 3rd law

$$\frac{T^2}{r^3} = k \quad \text{or} \quad \frac{r^3}{T^2} = \frac{1}{k}$$

$$\therefore F = \frac{4 m \pi^2}{r^2 K}$$

$$\text{But } \frac{4 m \pi^2}{K} = \text{Constant}$$

$$\therefore F = \text{constant} \times \frac{1}{r^2}$$

$$\therefore F \propto \frac{1}{r^2}$$

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Thus, Newton concluded that the centripetal force which is the force acting on the planet and is responsible for its circular motion, must be inversely proportional to the square of the distance between the planet and the Sun.

- d. A stone thrown vertically upwards with initial velocity u reaches a height 'h' before coming down. Show that the time taken to go up is same as the time taken to come down.

When stone is thrown up.

Initial velocity = $u \text{ m/s}$

Final velocity (v) = 0 m/s

Acceleration = $-g \text{ m/s}^2$

Time taken to go up = t_1

According to 1st eq. of motion

$$v = u + gt$$

$$0 = u - gt_1$$

$$gt_1 = u$$

$$t_1 = \frac{u}{g} \text{ seconds} \quad \dots (i)$$

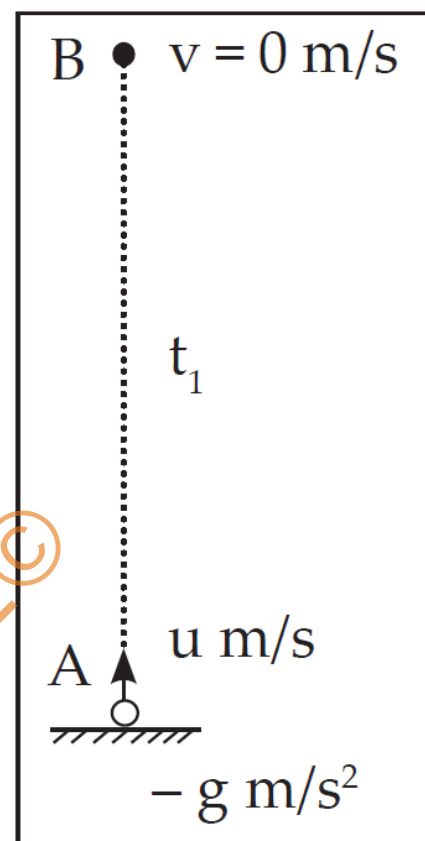
When stone falls down.

Initial velocity (u) = 0 m/s

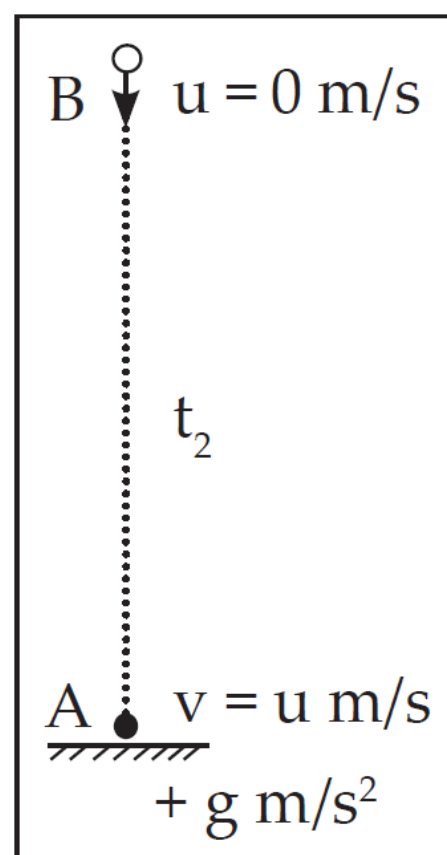
Final velocity (v) = $u \text{ m/s}$

Acceleration = $+g \text{ m/s}^2$

Time taken to go down = t_2



(During Ascent)



(During Descent)

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reference.
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According to 1st eq. of motion

$$v = u + gt$$

$$u = 0 + gt_2$$

$$t_2 = \frac{u}{g} \text{ seconds } \dots (ii)$$

From (i) and (ii),

$$t_1 = t_2$$

e. If the value of g suddenly becomes twice its value, it will become two times more difficult to pull a heavy object along the floor.

Why?

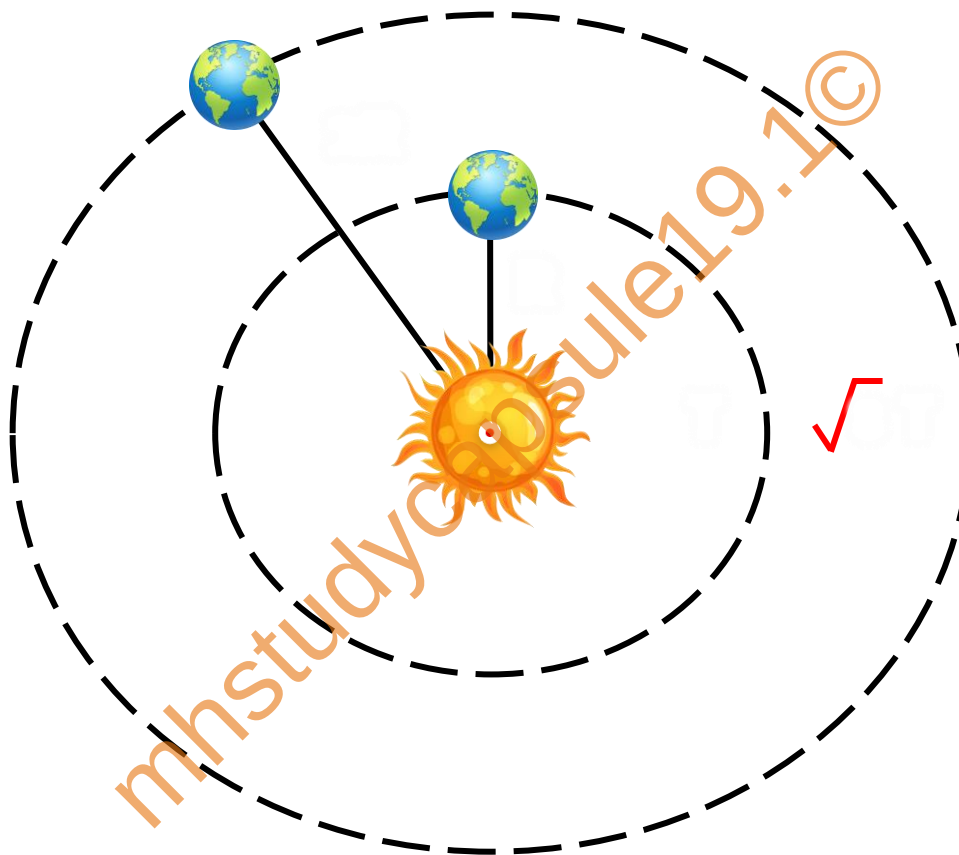
- i. If the value of g becomes double, the force with which the earth pulls the object i.e. weight of object becomes double.
- ii. As weight increases, frictional force also increases while pulling the object.
- iii. Hence, it will become two times more difficult to pull a heavy object along the floor.

3. Explain why the value of g is zero at the centre of the earth.

- i. At the centre of Earth, the force due to upper half of the Earth will cancel the force due to lower half.
- ii. In a similar manner, force due to any portion of the Earth will be cancelled due to the portion opposite to it.

- iii. Thus, the gravitational force at the centre on any body will be 0.
- iv. Since, from Newton's law, we know that $F = mg$.
- v. Since, mass m of an object can never be 0. Therefore, when $F = 0$, g must be 0. Thus, the value of g is zero at the centre of Earth.

4. Let the period of revolution of a planet at a distance R from a star be T . Prove that if it was at a distance of $2R$ from the star, its period of revolution will be $\sqrt{8} T$.



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Given:

Distance from Sun = R

Time of Rotation = T

New distance = $2R$

To find:

New time $T_N = ?$

Formula:

$$\frac{T^2}{R^3} = k$$

Solution:

Case (i)

$$\frac{T^2}{R^3} = k \quad \dots (i)$$

Case (ii)

$$\frac{T_N^2}{(2R)^3} = k \quad \text{OR} \quad \frac{T_N^2}{8R^3} = k \dots (ii)$$

From (i) and (ii),

$$\frac{T_N^2}{8R^3} = \frac{T^2}{R^3}$$

$$T_N^2 = 8T^2$$

$$T_N = \sqrt{8T^2}$$

$$T_N = \sqrt{8} T$$

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INTEXT QUESTIONS

1. What are the effects of force acting on an object?

The applied force can:

- i. Change the state of rest.
- ii. Change the direction of motion.
- iii. Change the speed of a moving object.
- iv. Change the shape of the object temporarily.

2. What types of forces are you familiar with?

Types of forces are:

- i. Gravitational force
- ii. Electro-magnetic force
- iii. Nuclear force
- iv. Frictional force

3. What do you know about gravitational force?

Every object in the universe attracts every other object with a force.

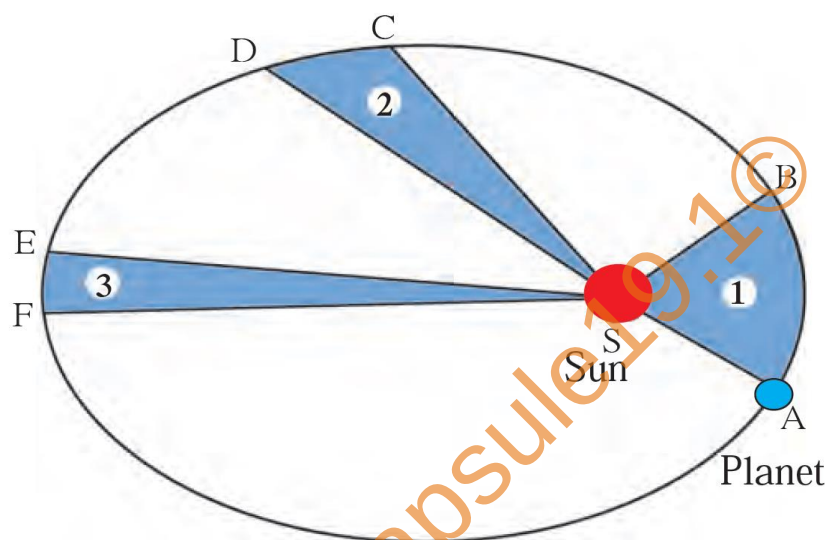
This force is called gravitational force.

4. What are Newton's laws of motion?

- i. Newton's first law of motion : Every inanimate object continues to be in a state of rest or of uniform motion in a straight line unless it is acted upon by an external unbalanced force.

- ii. Newton's second law of motion : The rate of change of momentum is directly proportional to the applied force and it takes place in the direction of force.
- iii. Newton's third law of motion : To every action there is always instantaneous, equal and opposite reaction.

5. If the area ESF in the figure is equal to area ASB, what will you infer about EF?



- i. Distance EF is lesser than AB. Hence, speed will be lesser at EF and greater at AB.
- ii. Therefore, in an elliptical orbit, speed is not constant.

6. Show that in SI units, the unit of G is Newton m² kg⁻².

As per universal law of gravitation,

$$F = \frac{G M m}{d^2}$$

Where:

F: Gravitational force, whose SI unit is Newton (N)

d: Distance between the masses, whose SI unit is metre (m)

M, m: Masses whose SI unit is kg.

Therefore, the SI unit of G is: Nm²kg⁻².

$$\therefore G = \frac{F d^2}{M m}$$

7. Is there a gravitational force between two objects kept on a table or between you and your friend sitting next to you? If yes, why don't the two move towards each other?

- i. Yes, there exists a gravitational force between me and my friend sitting beside each other.
- ii. The gravitational force between any two objects is given by, $F = G \frac{m_1 m_2}{d^2}$
- iii. Where, G = universal gravitational constant, m_1 and m_2 = mass of the two objects, d = distance between centers of the two objects.
- iv. Thus, me and my friend attract each other. But due to our small masses, we exert a force on each other, which is too small as compared to the gravitational force of the earth. Hence, me and my friend don't move towards each other.
- v. Apart from the gravitational force of the earth, there is frictional force acting on both me and my friend.

8. According to Newton's law of gravitation, every object attracts every other object. Thus, if the earth attracts an apple towards itself, the apple also attracts the earth towards itself with the same force. Why then does the apple fall towards the earth, but the earth does not move towards the apple?



- i. Yes, the earth also attracts the apple.
- ii. The apple has a small mass and the Earth has a large mass.
- iii. We know, $F = m a$ or $a = \frac{F}{m}$
- iv. Acceleration and mass are inversely proportional here.
- v. Therefore, the gravitational force between the two produces a large acceleration on the apple which has small mass.
- vi. The same gravitational force however produces a small (negligible) acceleration in the earth due to its large mass. This results in almost zero displacement of the earth.
- vii. Hence, we see the apple moving towards the earth and not the Earth moving towards the apple.

9. What would happen if there were no gravity?

- i. Without gravity, objects would not be held to the Earth's surface.
- ii. Everything would drift in space, and planets, stars, and galaxies would not be bound together in their orbits.

10. What would happen if the value of G was twice as large?

If the value of 'G' is doubled, then the gravitational force (F) also will get doubled.

11. What would be the value of g on the surface of the earth if its mass was twice as large and its radius half of what it is now?

Given: $M' = 2M$

PYQ

$$R' = \frac{1}{2} R = \frac{R}{2}$$

To find: $g' = ?$

Formula: $g = \frac{G M}{R^2}$

Solution: $g' = \frac{G M'}{(R')^2}$

$$g' = \frac{G \ 2M}{\left(\frac{R}{2}\right)^2}$$

$$g' = \frac{G \ 2M}{\frac{R^2}{4}}$$

$$g' = \frac{2 \times 4 \times G M}{R^2}$$

$$\therefore g = \frac{G M}{R^2}$$

$$g' = 2 \times 4 \times g$$

$$g' = 8g$$

The value of g will become 8 times.

12. Will the direction of the gravitational force change as we go inside the earth?

The direction of earth's gravitational force is towards the center of earth so it will not change.

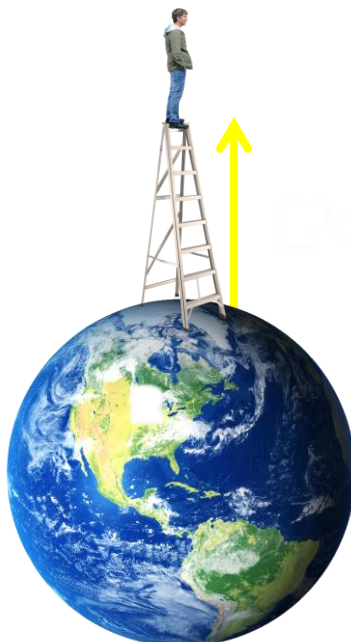
13. What will be the value of g at the centre of the earth?

At the centre of the Earth the value of ' g ' is zero.

14. Will your weight remain constant as you go above the surface of the earth?

- i. Weight of a body depends on acceleration due to gravity, $W = m \times g$.
- ii. As we go above the surface of the earth height increases and hence acceleration due to gravity decreases.
- iii. As acceleration due to gravity decreases weight will also decrease.
- iv. So, weight does not remain constant as we go above the surface of the earth.

15. Suppose you are standing on a tall ladder. If your distance from the centre of the earth is $2R$, what will be your weight?



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Given: $R' = 2R$

To find: $W' = ?$

Formula: $W = F = \frac{GMm}{R^2}$

Solution: $W = F = \frac{GMm}{R^2}$

$$W' = \frac{GMm}{(R')^2}$$

$$W' = \frac{GMm}{(2R)^2}$$

$$W' = \frac{GMm}{4R^2}$$

$$W' = \frac{1}{4} \left(\frac{GMm}{R^2} \right) \quad \because \quad W = \frac{GMm}{R^2}$$

$$W' = \frac{1}{4} W$$

Weight will be one fourth the original weight.

16. According to Newton's law of gravitation, earth's gravitational force is higher on an object of larger mass. Why doesn't that object fall down with higher velocity as compared to an object with lower mass?

- i. The acceleration due to gravity (g) on an object only depends on mass (M) and radius (R) of the earth i.e., $g = GM / R^2$.
- ii. It does not depend on mass (m) of the object.
- iii. The acceleration produced at any given point is the same for all objects.
- iv. Hence, an object having larger mass does not fall down with higher velocity as compared to an object with lower mass.

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Previous Year Questions

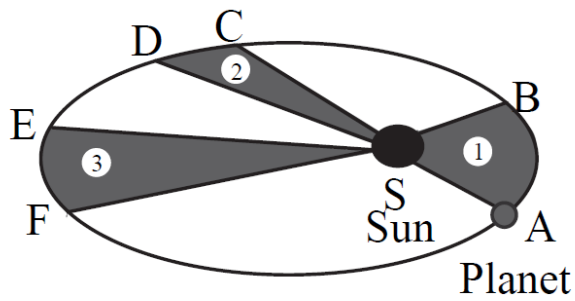
1. Explain why value of g changes if we go inside the earth.

- i. The value of g changes if we go inside the earth.
- ii. The value of r in equation $g = GM/R^2$ decreases and one would think that the value of g should increase as per the formula.
- iii. However, the part of the earth which contributes towards the gravitational force felt by the object also decreases.
- iv. Which means that the value of M to be used in equation $g = GM/R^2$ also decreases.
- v. As a combined result of change in r and M , the value of g decreases as we go deep inside the earth.

2. The weight of an object changes from place to place though its mass is constant.

- i. Mass is the amount of matter present in the object. Its value is same everywhere. Its value does not change even when we go to another planet.
- ii. The weight of an object is defined as the force with which the earth attracts the object. Weight, $W = F = m g$.
- iii. As the value of g is not same everywhere, the weight of an object changes from place to place, though its mass is constant everywhere.

3. Observe the given figure showing the orbit of a planet moving around the Sun and write the three laws related to it:



The orbit of a planet moving around the Sun

Exercise Question 2 - c

4. Distinguish between Mass and Weight.

Exercise Question 2 - a

P.T.O